

Collision Prevention in Diminished Reality through the Use of Peripheral Vision

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ABSTRACT

Diminished reality (DR) removes virtual and real objects from the user's view, decluttering and optimizing what the user sees through augmented/mixed reality. However, removing real objects create a potential safety concern because users may unintentionally bump into the diminished object. To address this issue, we apply characteristics of peripheral vision to DR and diminish objects when they are in the peripheral region and show the object when the user focuses on the object. We created 3 different object opacity control functions that use gaze information and evaluated them.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; **Virtual reality**.

KEYWORDS

Diminished Reality; Peripheral Vision; Opacity Control; Work Efficiency

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1 INTRODUCTION

The popularization of extended reality head-mounted displays (HMDs) is now introducing extended reality to different domains once thought infeasible by not only augmenting but also diminishing reality by removing objects. This concept of removal is called diminished reality (DR) [3, 8]. DR declutters users' views, decreases distraction, and improves work efficiency [6]. However, DR is a developing technology with caveats for its implementation. To study

interactions in DR, previous works used VR [2, 3, 7, 13] and revealed how unintended collisions with diminished objects create a challenge [3].

To prevent collisions, different approaches were used to keep users aware of diminished objects. Cheng et al. used an outline effect to prevent collisions [3], while Yokoro et al. suggested grayscale and semi-transparent objects [13]. Although their research showed that outline effects did prevent collisions, the transparent DR effect was the most effective with decluttering [13]. This causes a dilemma as transparent objects most effectively serve the main purpose of DR, but methods to prevent unintended collisions reduce the benefits.

To address this problem, we apply peripheral vision to DR. Peripheral vision is the area of vision outside the fovea [12]. The characteristic of objects being indistinguishable in this region is used to design a foveated rendering system in computer graphics [10, 11]. Other applications of peripheral vision include displaying virtual objects [1] or annotations in AR [5].

We propose a novel method of automatically uncovering hidden objects called peripheral diminished reality (PDR). PDR employs method in gaze-based technologies to calculate which area the user is focusing on. In PDR, when the user's fovea is near a diminished object, the object will automatically be shown, assuming that task-relevant objects will be placed inside the fovea. When it exits the area back into the peripheral vision, the object will become transparent.

The contribution of our work is the following: (1) The exploration of diminished objects in peripheral vision, mainly the optimal opacity control function. (2) The implementation of a novel concept of PDR. (3) Evaluation to assess performance and comfort of PDR by comparing different opacity control methods.

2 PDR IMPLEMENTATION

To implement PDR, we use gaze information from our HMD. This information is called by each object in VR to calculate the gaze's relative position with the object and to detect collision with the ray. We use these calculations in our opacity control functions, which are used to control objects' opacity.

2.1 Hardware

We used Varjo XR-3 for our HMD and its integrated 200 Hz eye-tracking feature to track the users' gaze. We also used HTC Valve Index controllers for scene navigation. We developed PDR using

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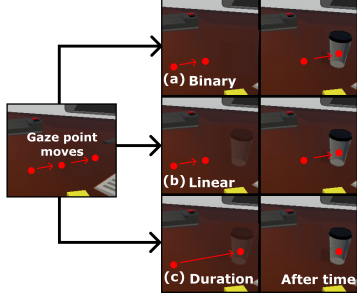


Figure 1: Implementation of opacity control functions: (a) Binary, (b) Linear, and (c) Duration. The red dot and arrow simulate the movement of the user’s gaze point.

C# on a Unity 3D Engine (2021.3.22f1). This setup was run on a desktop PC with a 3.00 GHz Intel Core i9 processor and NVIDIA RTX 4070Ti.

2.2 Opacity Control Functions

We develop three functions for object opacity control: *Binary*, *Linear*, and *Duration*. (Figure 1) While there are other possible opacity control function shapes, we selected these specific three to focus more on the simplest form of PDR rather than the details such as function shape.

The *Binary* function conveys the most basic form of PDR, revealing objects immediately when users are looking at them. When the distance between the ray and the object is below a threshold, we determined that the user is looking at the object, and the opacity was changed from 0 to 1.

The *Linear* function focuses on using distance as the main parameter. The function gradually increases the opacity of the object as the users’ gaze approaches it, while decrease opacity when the gaze moves away.

The *Duration* function uses gaze fixation time along with distance as its main parameter. Similar to the *Linear*, the *Duration* function increases opacity based on distance, but is fixed to a smaller value until the user fixates on the object long enough.

3 OPACITY CONTROL FUNCTION STUDY

We first explore the optimal function for opacity control in PDR. We compare our 3 different functions to obtain the best function for avoiding unintended collisions. The task of the study is to pick and place objects that accompany physical movements that may cause unintended collisions.

3.1 Procedure

We recruited 6 participants (3 male, 3 female; Age: $M = 25.33$, $SD = 2.16$). Participants received a short explanation of the task and the features of PDR, but the specifics such as opacity function were not mentioned. After the explanation, participants put on the HMD and used the controller, then entered the VR scene.

The scene contained a desk with 20 different objects used in daily life. To prevent participants from memorizing the scenes, we created 3 object arrangements by changing objects and their locations.

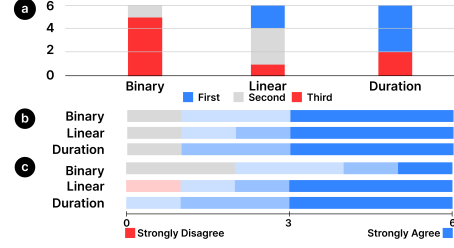


Figure 2: Results of opacity control function study: (a) preference ranking, (b) comfort, and (c) perceived performance.

Different opacity functions were applied for each arrangement. In each arrangement, we pre-selected 10 objects applied with PDR. Out of the remaining objects not applied with PDR, we indicated 5 objects that users had to retrieve. When participants’ hands or the retrieval target object collided with other objects, the controller vibrated to indicate a collision.

At the end of each opacity function, we measured the total number of collisions and completion time. We also asked participants a 7-point Likert scale questionnaire on comfort and perceived performance. After completing all PDR opacity functions, participants ranked the functions by preference.

3.2 Results

Linear function had the shortest average completion time (34.27 seconds), followed by *Duration* and *Binary*. For average collisions, *Linear* function again obtained the lowest average collisions (5.83), but this time followed by *Binary* then *Duration*.

The higher completion time for *Binary* function may be due to its disorientating nature as objects would blink too rapidly. This is supported by prior research on eye gaze that reveals how gaze is drawn to visual motion and saliency [4, 9]. This characteristics in *Binary* resulted in higher distractions in our user study. “*Binary changed too suddenly.*” (P1). However, *Binary* still showed less average collision counts compared to *Duration*, providing evidence that fully opaque objects are crucial for keeping users aware of potential obstacles.

In both comfort and perceived performance, *Duration* function received the highest average scores (6.17 and 6.33). *Linear* and *Binary* followed in order. As for preference, participants preferred *Duration* most frequently.

From these results, it is clear that participants prefer a middle ground when transitioning an object’s opacity from transparent to opaque. “*Both Linear and Duration were good because they didn’t immediately show the object fully.*” (P4). *Binary* function changed opacity too extremely, which is why it performed poorly in qualitative measures despite providing full information for object location.

Duration function’s conflicting quantitative and qualitative performances may be caused by the parameter values. “*I did not know Duration became fully transparent when I stared at it long enough.*” (P4). Due to the lack of time, the parameters were chosen through a short pilot study. However, the results of the opacity control function study emphasize a need for a more rigorous exploration of parameters to better satisfy users.

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